SKW

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Chapter 1

Setup

1.1 The cyclotomic field K

Let p be a prime number and f a positive integer. Set $q = p^f$. Let $K = \mathbb{Q}(\zeta_{q-1})$ be the $(q-1)$-th cyclotomic field. We fix a maximal ideal P of $\mathcal{O}_K$ lying above p.

1.2 The Teichmüller character

Definition 1. The *Teichmüller character* is the unique group isomorphism

$$\omega : (\mathcal{O}_K/P)^\times \xrightarrow{\sim} \mu_{q-1}(\mathcal{O}_K)$$

satisfying $\omega(x) \equiv x \pmod{P}$ for all $x \in (\mathcal{O}_K/P)^\times$, extended to all of $\mathcal{O}_K/P$ by $\omega(0) = 0$.

Lemma 2. *The order of ω as a multiplicative character is $q-1$.*

Proof. Since ω is an isomorphism between groups of order $q-1$, it has order $q-1$ as a group homomorphism. □

Lemma 3. *Let R be a ring containing $\mathcal{O}_K$ and let σ be a ring endomorphism of R such that $\sigma(\zeta_{q-1}) = \zeta_{q-1}^m$ for some $m \in \mathbb{Z}$. Then $\sigma(\omega(x)) = \omega(x)^m$ for all $x \in \mathcal{O}_K/P$.*

Proof. Since $\omega(x) \in \mu_{q-1}(\mathcal{O}_K)$, we have $\omega(x) = \zeta_{q-1}^k$ for some k. Then $\sigma(\omega(x)) = \sigma(\zeta_{q-1})^k = \zeta_{q-1}^{mk} = \omega(x)^m$. □

Lemma 4. *For all $a \in \mathbb{Z}$ with $(q-1) \nmid a$,*

$$\sum_{x \in \mathcal{O}_K/P} \omega(x)^{-a} = 0.$$

Proof. Since $(q-1) \nmid a$, the character ω^{-a} is nontrivial by 2. By `MulChar.sum_eq_zero_of_ne_one`, $\sum_{x\in\mathcal{O}_K/P} \omega(x)^{-a} = 0$. □

1.3 The additive character

Let L be an extension of K containing a primitive p-th root of unity ζ_p. We fix such a $\zeta_p \in \mathcal{O}_L$.

Definition 5. The *additive character* $\psi : \mathcal{O}_K/P \to \mu_p(\mathcal{O}_L)$ is defined by

$$\psi(x) = \zeta_p^{\mathrm{tr}_{(\mathcal{O}_K/P)/(\mathbb{Z}/p\mathbb{Z})}(x)}$$

where the exponent is lifted canonically to $\mathbb{Z}$.

Lemma 6. *The character ψ is nontrivial, i.e., $\psi \neq 1$.*

Proof. Since the extension $\mathcal{O}_K/P$ over $\mathbb{Z}/p\mathbb{Z}$ is separable, the trace $\mathrm{tr} : \mathcal{O}_K/P \to \mathbb{Z}/p\mathbb{Z}$ is nonzero, so there exists x with $\mathrm{tr}(x) \neq 0$, hence $\psi(x) = \zeta_p^{\mathrm{tr}(x)} \neq 1$. □

Lemma 7. *The character ψ is primitive.*

Proof. Since $\mathcal{O}_K/P$ is a field, any nontrivial additive character is primitive. The character ψ is nontrivial by 6. □

Lemma 8. *For all $x \in \mathcal{O}_K/P$, $\psi(x^p) = \psi(x)$.*

Proof. The Frobenius map $\varphi : x \mapsto x^p$ is a $(\mathbb{Z}/p\mathbb{Z})$-algebra automorphism of $\mathcal{O}_K/P$. By `Algebra.trace_eq_of_algEquiv`, $\mathrm{tr}(\varphi(x)) = \mathrm{tr}(x)$, i.e., $\mathrm{tr}(x^p) = \mathrm{tr}(x)$. Therefore $\psi(x^p) = \zeta_p^{\mathrm{tr}(x^p)} = \zeta_p^{\mathrm{tr}(x)} = \psi(x)$. □

Lemma 9. *Let $\mathfrak{P}$ be a prime ideal of $\mathcal{O}_L$ with $\zeta_p - 1 \in \mathfrak{P}$. Then*

$$\psi(x) \equiv 1 + \mathrm{tr}(x) \cdot (\zeta_p - 1) \pmod{\mathfrak{P}^2}$$

for all $x \in \mathcal{O}_K/P$.

Proof. There exists $a \in \mathbb{N}$ such that $\mathrm{tr}(x) = a$ and $\psi(x) = \zeta_p^a$. By the binomial theorem,

$$\psi(x) = \zeta_p^a = (1 + (\zeta_p - 1))^a = \sum_{n=0}^{a} \binom{a}{n} (\zeta_p - 1)^n.$$

Since $\zeta_p - 1 \in \mathfrak{P}$, we have $(\zeta_p - 1)^n \in \mathfrak{P}^n \subseteq \mathfrak{P}^2$ for all $n \geq 2$. Therefore

$$\psi(x) \equiv 1 + a(\zeta_p - 1) = 1 + \mathrm{tr}(x) \cdot (\zeta_p - 1) \pmod{\mathfrak{P}^2}.$$

□

Lemma 10. *Let $\mathfrak{P}$ be a prime ideal of $\mathcal{O}_L$ with $\zeta_p - 1 \in \mathfrak{P}$. Then $\psi(x) \equiv 1 \pmod{\mathfrak{P}}$ for all $x \in \mathcal{O}_K/P$.*

Proof. By 9, $\psi(x) \equiv 1 + \mathrm{tr}(x) \cdot (\zeta_p - 1) \pmod{\mathfrak{P}^2}$. Since $\zeta_p - 1 \in \mathfrak{P}$, the term $\mathrm{tr}(x) \cdot (\zeta_p - 1) \in \mathfrak{P}$, hence $\psi(x) \equiv 1 \pmod{\mathfrak{P}}$. □

Chapter 2

The Gauss sum

Throughout this chapter and the next two, we assume that p is an odd prime.

Let $\mathfrak{P}$ be a prime ideal of $\mathcal{O}_L$ with $\zeta_p - 1 \in \mathfrak{P}$. Let F be the subfield of L with $F = \mathbb{Q}(\zeta_p)$. The Galois group $\mathrm{Gal}(L/F)$ is isomorphic to $(\mathbb{Z}/(q-1)\mathbb{Z})^\times$ via the map $\sigma \mapsto a$ where $\sigma(\zeta_{q-1}) = \zeta_{q-1}^a$. We denote by $\mathfrak{n}(\sigma) \in \{1, \ldots, q-2\}$ the natural number corresponding to σ under this isomorphism.

2.1 Gauss sums

Definition 11. For $a \in \mathbb{Z}$, the *Gauss sum* is

$$g_a = g(\omega^{-a}, \psi) = \sum_{x \in \mathcal{O}_K/P} \omega(x)^{-a} \cdot \psi(x).$$

Lemma 12. *If $(q-1) \mid k$, then for all $a \in \mathbb{Z}$, $g_{a+k} = g_a$.*

Proof. Since $\omega^{q-1} = 1$ by 2, $(q-1) \mid k$ implies $\omega^{-k} = 1$, so $\omega^{-(a+k)} = \omega^{-a}$, hence $g_{a+k} = g_a$. ☐

Lemma 13. *If $(q-1) \nmid a$, then $g_a \in \mathfrak{P}$.*

Proof. By 10, $\psi(x) \equiv 1 \pmod{\mathfrak{P}}$ for all $x \in \mathcal{O}_K/P$. Therefore

$$g_a = \sum_{x \in \mathcal{O}_K/P} \omega(x)^{-a}\psi(x) \equiv \sum_{x \in \mathcal{O}_K/P} \omega(x)^{-a} \pmod{\mathfrak{P}}.$$

Since $(q-1) \nmid a$, we have $\omega^{-a} \neq 1$ by 2. Hence $\sum_{x \in \mathcal{O}_K/P} \omega(x)^{-a} = 0$ by 4. Therefore $g_a \equiv 0 \pmod{\mathfrak{P}}$, i.e. $g_a \in \mathfrak{P}$. ☐

Lemma 14. *If $(q-1) \nmid a$, then*

$$g_a \cdot g_{-a} = \omega^a(-1) \cdot p^f.$$

Proof. Since $(q-1) \nmid a$, ω^{-a} is nontrivial and ψ is primitive by 7. By `gaussSum_mul_gaussSum_eq_card`, $g(\omega^{-a}, \psi) \cdot g(\omega^a, \psi^{-1}) = p^f$. By the change of variable $x \mapsto -x$, $g(\omega^a, \psi^{-1}) = \omega^a(-1) \cdot g_{-a}$. Hence $g_a \cdot g_{-a} = \omega^a(-1) \cdot p^f$. ☐

Lemma 15. *If $(q-1) \nmid a$, then there exists $k \geq 1$ such that $N_{L/\mathbb{Q}}(g_a) = \pm p^k$. In particular, the only rational prime dividing $N_{L/\mathbb{Q}}(g_a)$ is p.*

Proof. By 14, $g_a \cdot g_{-a} = \omega^a(-1) \cdot p^f = \pm p^f$. Taking norms, $N_{L/\mathbb{Q}}(g_a) \cdot N_{L/\mathbb{Q}}(g_{-a}) = \pm p^{f \cdot [L:\mathbb{Q}]}$. Since both norms are integers and their product is $\pm p^{f \cdot [L:\mathbb{Q}]}$, each is of the form $\pm p^k$ for some $k \geq 0$. By 13, $g_a \in \mathfrak{P}$, so g_a is not a unit in $\mathcal{O}_L$, hence $N_{L/\mathbb{Q}}(g_a) \neq \pm 1$, thus $k \geq 1$. $\square$

Lemma 16. *Assume $\zeta_p - 1 \in \mathfrak{P}$. Then*

$$g_1 \equiv -(\zeta_p - 1) \pmod{\mathfrak{P}^2}.$$

Proof. By 9, $\psi(x) \equiv 1 + \mathrm{tr}(x) \cdot (\zeta_p - 1) \pmod{\mathfrak{P}^2}$ for all $x \in \mathcal{O}_K/P$. Therefore

$$g_1 \equiv \sum_x \omega(x)^{-1} + (\zeta_p - 1) \sum_x \omega(x)^{-1}\mathrm{tr}(x) \pmod{\mathfrak{P}^2}.$$

Since p is odd, $q - 1 \geq 2$, so $\omega^{-1} \neq 1$ by 2, and $\sum_x \omega(x)^{-1} = 0$ by orthogonality of characters. Moreover, Computing the trace via powers of Frobenius, $\sum_{x \in \mathcal{O}_K/P} \omega(x)^{-1} \cdot \mathrm{tr}(x) = -1$. Hence $g_1 \equiv -(\zeta_p - 1) \pmod{\mathfrak{P}^2}$. $\square$

Lemma 17. *The extension L/F is Galois with $\mathrm{Gal}(L/F) \cong (\mathbb{Z}/(q-1)\mathbb{Z})^\times$, via the action on ζ_{q-1}.*

Proof. Since $\gcd(q-1, p) = 1$, the fields $K = \mathbb{Q}(\zeta_{q-1})$ and $F = \mathbb{Q}(\zeta_p)$ are linearly disjoint over $\mathbb{Q}$. Therefore $L = KF$ is Galois over F with $\mathrm{Gal}(L/F) \cong \mathrm{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/(q-1)\mathbb{Z})^\times$. $\square$

Lemma 18. *For all $\sigma \in \mathrm{Gal}(L/F)$ and $a \in \mathbb{Z}$,*

$$\sigma(g_a) = g_{a \cdot \mathfrak{n}(\sigma)}.$$

Proof. Since $\sigma \in \mathrm{Gal}(L/F)$, it fixes ζ_p, hence ψ. By 3, $\sigma(\omega(x)) = \omega(x)^{\mathfrak{n}(\sigma)}$ for all x. Therefore

$$\sigma(g_a) = \sum_{x \in \mathcal{O}_K/P} \sigma(\omega(x)^{-a}) \cdot \psi(x) = \sum_{x \in \mathcal{O}_K/P} \omega(x)^{-a \cdot \mathfrak{n}(\sigma)} \cdot \psi(x) = g_{a \cdot \mathfrak{n}(\sigma)}.$$

$\square$

Lemma 19. *For all $a \in \mathbb{Z}$, $g_{pa} = g_a$.*

Proof. The Frobenius map $\varphi : x \mapsto x^p$ is a bijection on $\mathcal{O}_K/P$. Using the substitution $y = \varphi(x)$,

$$g_{pa} = \sum_{x \in \mathcal{O}_K/P} \omega(x)^{-pa}\psi(x) = \sum_{y \in \mathcal{O}_K/P} \omega(\varphi^{-1}(y))^{-pa}\psi(\varphi^{-1}(y)).$$

By multiplicativity of ω, $\omega(\varphi^{-1}(y))^p = \omega(\varphi^{-1}(y)^p) = \omega(y)$, so $\omega(\varphi^{-1}(y))^{-pa} = \omega(y)^{-a}$. By 8, $\psi(\varphi^{-1}(y)) = \psi(y)$. Therefore $g_{pa} = g_a$. $\square$

2.2 Jacobi sums

Lemma 20. *If $(q-1) \nmid (a+b)$, then*

$$g_a \cdot g_b = g_{a+b} \cdot J(\omega^{-a}, \omega^{-b}).$$

Proof. This follows from `jacobiSum_mul_nontrivial` applied to $\chi = \omega^{-a}$ and $\phi = \omega^{-b}$. $\square$

Lemma 21. *For all $a, b \in \mathbb{Z}$, $J(\omega^{-a}, \omega^{-b}) \in \mathcal{O}_K$.*

Proof. $\square$

Chapter 3

The valuation function s

Definition 22. For $a \in \mathbb{Z}$, we define

$$s(a) = v_{\mathfrak{P}}(g_a)$$

Lemma 23. *If* $(q-1) \mid a$, *then* $s(a) = 0$.

Proof. Since $(q-1) \mid a$, $\omega^{-a} = 1$ by 2, so $g_a = \sum_x \psi(x)$, which equals 0 by 68 since ψ is nontrivial by 6. Hence $s(a) = v_{\mathfrak{P}}(0) = 0$. $\square$

Lemma 24. *For all* $a, b \in \mathbb{Z}$, $s(a+b) \leq s(a) + s(b)$.

Proof. If $(q-1) \mid (a+b)$ then $s(a+b) = 0$ by 23 and the result is trivial. Otherwise, by 20,

$$g_a \cdot g_b = g_{a+b} \cdot J(\omega^{-a}, \omega^{-b}).$$

Taking valuations and using the fact that $J(\omega^{-a}, \omega^{-b}) \in \mathcal{O}_L$, hence $v_{\mathfrak{P}}(J) \geq 0$, we obtain $s(a) + s(b) \geq s(a+b)$. $\square$

Lemma 25. *Assume* $(q-1) \mid k$. *Then*

$$s(a) + s(k-a) = \begin{cases} 0 & \text{if } (q-1) \mid a \\ (p-1) \cdot f & \text{otherwise.} \end{cases}$$

Proof. Since $(q-1) \mid k$, we have $s(k-a) = s(-a)$ by periodicity of g: by 12, $g_{a+(q-1)} = g_a$, so s has period $q-1$.

If $(q-1) \mid a$, then $(q-1) \mid (-a)$, so $s(a) = s(-a) = 0$ by 23 and the result is clear.

If $(q-1) \nmid a$, by 14, $g_a \cdot g_{-a} = \omega^a(-1) \cdot p^f$. Since $\omega^a(-1)$ is a root of unity, it is a unit in $\mathcal{O}_L$, hence $v_{\mathfrak{P}}(\omega^a(-1)) = 0$. By 66, $v_{\mathfrak{P}}(p^f) = f \cdot e(\mathfrak{P} \mid p) = f(p-1)$. Therefore

$$s(a) + s(-a) = v_{\mathfrak{P}}(g_a) + v_{\mathfrak{P}}(g_{-a}) = v_{\mathfrak{P}}(g_a \cdot g_{-a}) = v_{\mathfrak{P}}(p^f) = (p-1) \cdot f.$$

$\square$

Lemma 26. *For all* $a \in \mathbb{Z}$, $s(pa) = s(a)$.

Proof. By 19, $g_{pa} = g_a$, hence $s(pa) = v_{\mathfrak{P}}(g_{pa}) = v_{\mathfrak{P}}(g_a) = s(a)$. $\square$

Lemma 27. $s(1) = 1$.

Proof. By 13, $g_1 \in \mathfrak{P}$, so $s(1) \geq 1$. By 16, $g_1 \equiv -(\zeta_p - 1) \pmod{\mathfrak{P}^2}$. Since $\zeta_p - 1 \notin \mathfrak{P}^2$ by 67, we have $g_1 \notin \mathfrak{P}^2$, so $s(1) \leq 1$. □

Lemma 28. *For all $a \in \mathbb{Z}$ with $a \geq 0$, $s(a) \leq a$.*

Proof. By induction on a. The case $a = 0$ follows from 23 (since $(q-1) \mid 0$). For $a \geq 1$, by 24 and the induction hypothesis,

$$s(a) \leq s(1) + s(a-1) \leq 1 + (a-1) = a.$$

□

Lemma 29. *Let $0 \leq a$ and let $a = a_0 + a_1 p + \cdots + a_{t-1}p^{t-1}$ be the p-adic expansion of a. Then*

$$s(a) \leq a_0 + a_1 + \cdots + a_{t-1}.$$

Proof. By induction on the list of digits $(a_0, \ldots, a_{t-1})$. The base case $a = 0$ is immediate. For the inductive step, write $a = a_0 + p \cdot a'$ where $a' = a_1 + a_2 p + \cdots + a_{t-1}p^{t-2}$. By 24,

$$s(a_0 + p \cdot a') \leq s(a_0) + s(p \cdot a').$$

By 28, $s(a_0) \leq a_0$. By 26, $s(p \cdot a') = s(a')$. By the induction hypothesis, $s(a') \leq a_1 + \cdots + a_{t-1}$. Hence $s(a) \leq a_0 + a_1 + \cdots + a_{t-1}$. □

Lemma 30. *We have*

$$\sum_{a=0}^{q-2} s(a) = \frac{(q-2) \cdot f \cdot (p-1)}{2}.$$

Proof. We pair each $a \in \{0, \ldots, q-2\}$ with $q-1-a$. By 25 with $k = q-1$, for any a with $(q-1) \nmid a$,

$$s(a) + s(q-1-a) = (p-1) \cdot f.$$

Note that $s(0) = 0$ and $s(q-1) = 0$ by 23.

Since p is odd, $q - 1 = p^f - 1$ is even. The set $\{1, \ldots, q-2\}$ consists of $(q-3)/2$ pairs $\{a, q-1-a\}$ with $a \neq q-1-a$, plus the central term $a = (q-1)/2$ which pairs with itself. Since $q \geq 3$, we have $1 \leq (q-1)/2 \leq q-2$, so $(q-1) \nmid (q-1)/2$, and 25 gives $2s((q-1)/2) = (p-1)f$, hence $s((q-1)/2) = (p-1)f/2$. Therefore

$$\sum_{a=0}^{q-2} s(a) = \frac{q-3}{2} \cdot (p-1)f + \frac{(p-1)f}{2} = \frac{(q-2)(p-1)f}{2}.$$

□

Proposition 31. *Let $0 \leq a < q-1$ and let $a = a_0 + a_1 p + \cdots + a_{f-1}p^{f-1}$, $0 \leq a_i \leq p-1$, be the p-adic expansion of a. Then*

$$s(a) = a_0 + a_1 + \cdots + a_{f-1}.$$

Proof. By 29, $s(a) \leq a_0 + \cdots + a_{f-1}$ for all a. Summing over $0 \leq a \leq q-2$, we get

$$\sum_{a=0}^{q-2} s(a) \leq \sum_{a=0}^{q-2} (a_0 + \cdots + a_{f-1}).$$

By `Nat.sum_sum_digits_eq`, the right-hand side equals $\frac{(q-2) \cdot f \cdot (p-1)}{2}$, which equals the left-hand side by 30. Since each term satisfies $s(a) \leq a_0 + \cdots + a_{f-1}$ and the sums are equal, we must have $s(a) = a_0 + \cdots + a_{f-1}$ for all a. □

Chapter 4

Ideal factorization of Gauss sums

4.1 Setup for Stickelberger's theorem

Let m be a positive integer dividing $q-1$ such that f is the multiplicative order of p modulo m. Let $d = (q-1)/m$.

Let $E = \mathbb{Q}(\zeta_m, \zeta_p)$. We denote by $\tilde{\mathfrak{p}}$ the prime ideal of $\mathcal{O}_E$ lying below $\mathfrak{P}$, i.e. $\tilde{\mathfrak{p}} = \mathfrak{P} \cap \mathcal{O}_E$, and by $\tilde{\mathfrak{p}}_0$ the prime ideal of $\mathcal{O}_{\mathbb{Q}(\zeta_m)}$ lying below $\mathfrak{P}$, i.e. $\tilde{\mathfrak{p}}_0 = \mathfrak{P} \cap \mathcal{O}_{\mathbb{Q}(\zeta_m)}$.

Let $G = \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$. For $(a,m) = 1$, let $\sigma_a \in G$ denote the element with $\sigma_a(\zeta_m) = \zeta_m^a$. The *Stickelberger element* is

$$\Theta = \Theta(\mathbb{Q}(\zeta_m)) = \sum_{\substack{a=1 \\ (a,m)=1}}^{m} a \cdot \sigma_a^{-1} \in \mathbb{Z}[G].$$

Remark 1. With the notation of Washington, $\Theta = m\theta$ where $\theta = \sum_{(a,m)=1} \{a/m\} \sigma_a^{-1} \in \mathbb{Q}[G]$.

4.2 Descent and Stickelberger's theorem

Lemma 32. *For $0 \leq h < q-1$,*

$$s(h) = (p-1) \sum_{i=0}^{f-1} \left\{ \frac{p^i h}{q-1} \right\}.$$

Proof. Let $h = a_0 + a_1 p + \cdots + a_{f-1} p^{f-1}$ be the p-adic expansion of h. By 31, $s(h) = a_0 + \cdots + a_{f-1}$. On the other hand, $p^i h \equiv a_0 p^i + \cdots + a_{f-1} p^{i+f-1} \pmod{q-1}$, so

$$\left\{ \frac{p^i h}{q-1} \right\} = \frac{1}{q-1}(a_0 p^i + a_1 p^{i+1} + \cdots + a_{f-1} p^{i-1}),$$

where the indices are taken mod f. Summing over $i = 0, \ldots, f-1$, each digit a_j appears multiplied by $(p^0 + p^1 + \cdots + p^{f-1})/(q-1) = 1/(p-1)$. Therefore

$$(p-1) \sum_{i=0}^{f-1} \left\{ \frac{p^i h}{q-1} \right\} = a_0 + \cdots + a_{f-1} = s(h).$$

$\square$

Lemma 33. *For $a \in (\mathbb{Z}/m\mathbb{Z})^\times$,*

$$m \cdot s(ad) = (p-1)\sum_{i=0}^{f-1} (ap^i \bmod m).$$

Proof. By 32 applied to $h = ad$,

$$s(ad) = (p-1)\sum_{i=0}^{f-1} \left\{ \frac{p^i \cdot ad}{q-1} \right\} = (p-1)\sum_{i=0}^{f-1} \left\{ \frac{ap^i}{m} \right\}.$$

Since $m \cdot \{ap^i/m\} = ap^i \bmod m$ for each i, multiplying by m gives the result. $\square$

Lemma 34. *For any $\tau \in \mathrm{Gal}(L/\mathbb{Q}(\zeta_m, \zeta_p))$, $\tau(g_d^m) = g_d^m$.*

Proof. Since τ fixes ζ_m, we have $\mathfrak{n}(\tau) \equiv 1 \pmod m$, so $d \cdot \mathfrak{n}(\tau) \equiv d \pmod{q-1}$. By 18, $\tau(g_d) = g_{d\cdot\mathfrak{n}(\tau)} = g_d$. Therefore $\tau(g_d^m) = g_d^m$. $\square$

Lemma 35. *For any $\tau \in \mathrm{Gal}(L/K)$ with $\tau(\zeta_p) = \zeta_p^e$ and $e \in (\mathbb{Z}/p\mathbb{Z})^\times$,*

$$\tau(g_d) = \omega^d(e)^{-1} \cdot g_d.$$

Proof. Since $\tau \in \mathrm{Gal}(L/K)$, it fixes $K = \mathbb{Q}(\zeta_{q-1})$, hence fixes all $(q-1)$-th roots of unity, hence fixes $\omega(x)$ for all x. Since $\tau(\zeta_p) = \zeta_p^e$, we have $\tau(\psi(x)) = \psi(ex)$. Therefore

$$\tau(g_d) = \sum_{x \in \mathcal{O}_K/P} \omega(x)^{-d}\psi(ex).$$

By the substitution $y = ex$ (bijection on $\mathcal{O}_K/P$ since $e \in (\mathbb{Z}/p\mathbb{Z})^\times \subset \mathbb{F}_q^\times$),

$$\sum_x \omega(x)^{-d}\psi(ex) = \sum_y \omega(y/e)^{-d}\psi(y) = \omega^d(e)^{-1}\sum_y \omega(y)^{-d}\psi(y) = \omega^d(e)^{-1} g_d.$$

$\square$

Lemma 36. *For any $\tau \in \mathrm{Gal}(L/K)$, $\tau(g_d^m) = g_d^m$.*

Proof. By 35, $\tau(g_d) = \omega^d(e)^{-1} g_d$ where e is such that $\tau(\zeta_p) = \zeta_p^e$. Since $md = q-1$, we have $\omega^d(e)^{-m} = \omega^{q-1}(e)^{-1} = 1$, hence $\tau(g_d^m) = \omega^d(e)^{-m} g_d^m = g_d^m$. $\square$

Lemma 37. *We have $g_d^m \in \mathcal{O}_{\mathbb{Q}(\zeta_m)}$.*

Proof. By 34, g_d^m is fixed by $\mathrm{Gal}(L/\mathbb{Q}(\zeta_m, \zeta_p))$, so $g_d^m \in \mathbb{Q}(\zeta_m, \zeta_p)$. By 36, g_d^m is fixed by $\mathrm{Gal}(L/K)$. Since $\mathbb{Q}(\zeta_m, \zeta_p) = KF$ and $\mathrm{Gal}(KF/\mathbb{Q}(\zeta_m))$ is generated by the restrictions of $\mathrm{Gal}(L/\mathbb{Q}(\zeta_m, \zeta_p))$ and $\mathrm{Gal}(L/K)$, we conclude $g_d^m \in \mathbb{Q}(\zeta_m)$. Since $g_d \in \mathcal{O}_L$, we have $g_d^m \in \mathcal{O}_{\mathbb{Q}(\zeta_m)}$. $\square$

Lemma 38. *The extension L/E is unramified above p.*

Proof. We have $L = E(\zeta_{q-1})$ and $E = \mathbb{Q}(\zeta_m, \zeta_p)$. Since $\gcd(q-1, p) = 1$, the extension $\mathbb{Q}(\zeta_{q-1})/\mathbb{Q}$ is unramified at p. Since the ramification above p in $L/\mathbb{Q}$ comes entirely from the adjunction of ζ_p, which is already in E, the extension L/E is unramified above p. $\square$

Lemma 39. *The extension $E/\mathbb{Q}(\zeta_m)$ is totally ramified above $\tilde{\mathfrak{p}}_0$ with ramification index $p-1$. In particular, $\tilde{\mathfrak{p}}^{p-1} = \tilde{\mathfrak{p}}_0 \cdot \mathcal{O}_E$.*

Proof. Since $\gcd(m,p) = 1$ (as $m \mid q-1$ and $\gcd(q-1,p) = 1$), the fields $\mathbb{Q}(\zeta_m)$ and $\mathbb{Q}(\zeta_p)$ are linearly disjoint over $\mathbb{Q}$, and $E = \mathbb{Q}(\zeta_m) \cdot \mathbb{Q}(\zeta_p)$. The extension $\mathbb{Q}(\zeta_p)/\mathbb{Q}$ is totally ramified at p with ramification index $p-1$. Since $\mathbb{Q}(\zeta_m)/\mathbb{Q}$ is unramified at p (as $p \nmid m$), the extension $E/\mathbb{Q}(\zeta_m)$ is totally ramified at $\tilde{\mathfrak{p}}_0$ with ramification index $p-1$. $\square$

Lemma 40. *The stabilizer of* $\tilde{\mathfrak{p}}_0$ *in* $G = \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ *is the subgroup* $\langle \sigma_p \rangle$*, which has order* f*. In particular, for* $a \in (\mathbb{Z}/m\mathbb{Z})^\times$ *and* $0 \leq i \leq f-1$*,*

$$\sigma_{ap^i}^{-1}(\tilde{\mathfrak{p}}_0) = \sigma_a^{-1}(\tilde{\mathfrak{p}}_0).$$

Proof. Since f is the multiplicative order of p modulo m by hypothesis, $\langle \sigma_p \rangle$ has order f in G. The stabilizer of $\tilde{\mathfrak{p}}_0$ in G is the decomposition group of $\tilde{\mathfrak{p}}_0$ over p, which is generated by σ_p since p generates the decomposition group. $\square$

Lemma 41. *For* $a \in (\mathbb{Z}/m\mathbb{Z})^\times$*,*

$$v_{\sigma_a^{-1}(\tilde{\mathfrak{p}}_0)}(g_d^m) = \sum_{i=0}^{f-1} (ap^i \bmod m).$$

Proof. Since $g_d \in \mathcal{O}_E$, we have

$$v_{\sigma_a^{-1}(\tilde{\mathfrak{p}})}(g_d) = v_{\tilde{\mathfrak{p}}}(\sigma_a(g_d)).$$

Lift $\sigma_a \in G = \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ to an element $\tilde{\sigma}_a \in \mathrm{Gal}(L/F)$ with $\mathfrak{n}(\tilde{\sigma}_a) \equiv a \pmod m$. By 18, $\tilde{\sigma}_a(g_d) = g_{d \cdot \mathfrak{n}(\tilde{\sigma}_a)}$. The ambiguity in the lift is an element of $\mathrm{Gal}(L/\mathbb{Q}(\zeta_m, \zeta_p))$, which acts on g_d by 35 as multiplication by a unit $\omega^d(e)^{-1} \in \mathcal{O}_E^\times$. Therefore $v_{\tilde{\mathfrak{p}}}(\sigma_a(g_d)) = v_{\tilde{\mathfrak{p}}}(g_{ad}) = s(ad)$, where the last equality uses that L/E is unramified above p by 38, so $v_{\tilde{\mathfrak{p}}}(g_{ad}) = v_{\mathfrak{P}}(g_{ad}) = s(ad)$. Since $E/\mathbb{Q}(\zeta_m)$ is totally ramified of degree $p-1$ above $\tilde{\mathfrak{p}}_0$ by 39, we have $v_{\sigma_a^{-1}(\tilde{\mathfrak{p}})} = (p-1) \cdot v_{\sigma_a^{-1}(\tilde{\mathfrak{p}}_0)}$. Therefore

$$v_{\sigma_a^{-1}(\tilde{\mathfrak{p}}_0)}(g_d^m) = \frac{m \cdot s(ad)}{p-1} = \sum_{i=0}^{f-1} (ap^i \bmod m),$$

where the last equality is 33. $\square$

Lemma 42. *For* $a \in (\mathbb{Z}/m\mathbb{Z})^\times$*,*

$$v_{\sigma_a^{-1}(\tilde{\mathfrak{p}}_0)}(\tilde{\mathfrak{p}}_0^\Theta) = \sum_{i=0}^{f-1} (ap^i \bmod m).$$

Proof. By definition of Θ and 40, regrouping the sum over $(\mathbb{Z}/m\mathbb{Z})^\times$ by orbits under $\langle p \rangle$:

$$v_{\sigma_a^{-1}(\tilde{\mathfrak{p}}_0)}(\tilde{\mathfrak{p}}_0^\Theta) = \sum_{b \in (\mathbb{Z}/m\mathbb{Z})^\times} b \cdot v_{\sigma_a^{-1}(\tilde{\mathfrak{p}}_0)}(\sigma_b^{-1}(\tilde{\mathfrak{p}}_0)) = \sum_{i=0}^{f-1} (ap^i \bmod m),$$

since $v_{\sigma_a^{-1}(\tilde{\mathfrak{p}}_0)}(\sigma_b^{-1}(\tilde{\mathfrak{p}}_0)) = 1$ if $b \equiv ap^i \pmod m$ for some i, and 0 otherwise. $\square$

Lemma 43.

$$(g_d^m) = \tilde{\mathfrak{p}}_0^\Theta$$

as ideals of $\mathcal{O}_{\mathbb{Q}(\zeta_m)}$*.*

Proof. By 37, $g_d^m \in \mathcal{O}_{\mathbb{Q}(\zeta_m)}$. By 15, g_d^m is only divisible by primes above p. Similarly, $\tilde{\mathfrak{p}}_0^\Theta$ only involves primes above p. It therefore suffices to check that for every prime $\mathfrak{q} = \sigma_a^{-1}(\tilde{\mathfrak{p}}_0)$ of $\mathcal{O}_{\mathbb{Q}(\zeta_m)}$ above p,

$$v_{\mathfrak{q}}(g_d^m) = v_{\mathfrak{q}}(\tilde{\mathfrak{p}}_0^\Theta).$$

By 41,

$$v_{\mathfrak{q}}(g_d^m) = \sum_{i=0}^{f-1} (ap^i \bmod m).$$

By 42,

$$v_{\mathfrak{q}}(\tilde{\mathfrak{p}}_0^\Theta) = \sum_{i=0}^{f-1} (ap^i \bmod m).$$

The two sides are equal, which completes the proof. □

Theorem 44. *The ideal $\tilde{\mathfrak{p}}_0^\Theta$ is principal in $\mathbb{Q}(\zeta_m)$.*

Proof. By 43, $(g_d^m) = \tilde{\mathfrak{p}}_0^\Theta$. □

Corollary 45. *The Stickelberger element Θ annihilates the ideal class group $\mathrm{Cl}(\mathbb{Q}(\zeta_m))$: for every ideal class c,*

$$c^\Theta = \prod_{\substack{a=1 \\ (a,m)=1}}^{m} \sigma_a^{-1}(c)^a = 1.$$

Proof. Every ideal class of a number field contains an integral ideal whose norm is coprime to any prescribed integer, so choose a representative $\mathfrak{b} \in c$ with $\gcd(N\mathfrak{b}, 2m) = 1$. Then $N\mathfrak{b}$ is odd and coprime to m, so applying 44 to each prime factor of $\mathfrak{b}$ and multiplying shows that $\mathfrak{b}^\Theta$ is principal. Hence $c^\Theta = [\mathfrak{b}]^\Theta = [\mathfrak{b}^\Theta] = 1$. □

Remark 2. Mathematically, coprimality of the representative to m alone suffices for Stickelberger's theorem. We additionally require the norm to be odd (i.e. coprimality to $2m$) only because the formalized form of 44 is proved for primes of odd residue characteristic.

Chapter 5

The Kronecker-Weber theorem

5.1 Reduction to the prime degree case

Throughout this chapter, ζ_m denotes a primitive m-th root of unity. We say that an extension is *unramified* if it is unramified at all finite primes. A normal extension K/F has *exponent* m if $\mathrm{Gal}(K/F)$ has exponent m.

The goal of this chapter is to prove the following theorem.

Theorem 46. *Every abelian extension of $\mathbb{Q}$ is contained in a cyclotomic field.*

Proof. By 49, it suffices to prove that every cyclic extension of $\mathbb{Q}$ of prime power degree p^m unramified outside p is contained in a cyclotomic field. For odd p, this is 55; for $p = 2$, this is 54. □

Lemma 47. *Every finite abelian extension $K/\mathbb{Q}$ is the compositum of cyclic extensions of prime power degree.*

Proof. The Galois group $\mathrm{Gal}(K/\mathbb{Q})$ is a finite abelian group, hence a direct product of cyclic groups of prime power order. Each factor corresponds to a cyclic subextension of prime power degree, and K is their compositum. □

Lemma 48. *Let $K/\mathbb{Q}$ be a cyclic extension of prime power degree p^m. Then there exists a cyclic extension $F/\mathbb{Q}$ of prime power degree p^m unramified outside p such that K is cyclotomic if and only if F is cyclotomic.*

Proof. The discriminant of K is a finite integer, so only finitely many primes $q \neq p$ are ramified in $K/\mathbb{Q}$. We apply 51 iteratively. At each step, given a cyclic degree-p^m extension $K'/\mathbb{Q}$ with some $q \neq p$ ramified, 51 provides a cyclotomic extension $\Lambda/\mathbb{Q}$ and a cyclic degree-p^m extension $K''/\mathbb{Q}$ in which q is unramified, such that $K'\Lambda = K''\Lambda$. Since Λ is cyclotomic, K' is cyclotomic if and only if K'' is (as $K'\Lambda = K''\Lambda$ and Λ is cyclotomic). Since $K'' \subseteq K'\Lambda$ and Λ is only ramified at q, the primes ramified in K'' are a subset of those of K' union $\{q\}$, which equals the primes of K'. Since q is unramified in K'' by construction, the set of primes $\neq p$ ramified in K'' is strictly smaller than that of K'. The process terminates with the desired F. □

Lemma 49. *It suffices to prove that every cyclic extension of $\mathbb{Q}$ of prime power degree unramified outside p is cyclotomic.*

Proof. By 47, any finite abelian extension of $\mathbb{Q}$ is a compositum of cyclic extensions of prime power degree. By 48, each such extension can be replaced by a cyclotomic extension times a cyclic extension of prime power degree unramified outside p. By 55 and 54, the latter is cyclotomic. $\square$

Lemma 50. *Let $L/\mathbb{Q}$ be a Galois extension with cyclic Galois group, and let K, K' be intermediate fields with $[K:\mathbb{Q}] \mid [K':\mathbb{Q}]$. Then $K \subseteq K'$.*

Proof. Via the Galois correspondence, $K \subseteq K'$ is equivalent to $\mathrm{Fix}(K') \leq \mathrm{Fix}(K)$ in $\mathrm{Gal}(L/\mathbb{Q})$. In a cyclic group, the subgroup lattice is totally ordered by divisibility of indices, and $[\mathrm{Gal}(L/\mathbb{Q}) : \mathrm{Fix}(K)] = [K:\mathbb{Q}]$ divides $[\mathrm{Gal}(L/\mathbb{Q}) : \mathrm{Fix}(K')] = [K':\mathbb{Q}]$, so $\mathrm{Fix}(K') \leq \mathrm{Fix}(K)$. $\square$

Lemma 51. *Let $K/\mathbb{Q}$ be a cyclic extension of prime power degree p^m. If $q \neq p$ is ramified in $K/\mathbb{Q}$, then there exists a cyclic cyclotomic extension $\Lambda/\mathbb{Q}$ and a cyclic extension $F/\mathbb{Q}$ of degree p^m in which q is unramified, such that $K\Lambda = F\Lambda$.*

Proof. Since $q \neq p$ and $[K:\mathbb{Q}] = p^m$, the prime q is tamely ramified in $K/\mathbb{Q}$. Let $e = p^s$ $(1 \leq s \leq m)$ be the ramification index of q in $K/\mathbb{Q}$; the inertia group $I_q \subseteq \mathrm{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/p^m\mathbb{Z}$ has order p^s.

Since $\phi(q^r) = q^{r-1}(q-1)$ and $\gcd(p,q) = 1$, for large enough r we have $p^s \mid \phi(q^r)$, so $\mathbb{Q}(\zeta_{q^r})$ contains a cyclic subfield Λ of degree p^s over $\mathbb{Q}$. This Λ is totally (tamely) ramified at q with ramification index p^s.

We may arrange $K \cap \Lambda = \mathbb{Q}$ by choosing r large: any nontrivial $K \cap \Lambda$ would be a p-extension ramified only at q (as K is a p-extension and Λ is unramified outside q), hence a subfield of the p-part of $\mathbb{Q}(\zeta_{q-1})$. Replacing Λ by a cyclic subfield of $\mathbb{Q}(\zeta_{q^r})$ lying above that p-part for large r ensures linear disjointness.

Set $A = K\Lambda$. With $K \cap \Lambda = \mathbb{Q}$, we have $\mathrm{Gal}(A/\mathbb{Q}) \cong \mathbb{Z}/p^m\mathbb{Z} \times \mathbb{Z}/p^s\mathbb{Z}$. Let $g \in \mathrm{Gal}(A/\mathbb{Q})$ be the element whose image in $\mathrm{Gal}(K/\mathbb{Q})$ is the generator σ of I_q and whose image in $\mathrm{Gal}(\Lambda/\mathbb{Q})$ is the generator τ of $\mathrm{Gal}(\Lambda/\mathbb{Q})$. Define $F = A^{\langle g \rangle}$.

- *Degree*: $[F:\mathbb{Q}] = |G|/|\langle g \rangle| = p^{m+s}/p^s = p^m$.
- *Cyclic*: $\mathrm{Gal}(F/\mathbb{Q}) \cong G/\langle g \rangle \cong (\mathbb{Z}/p^m\mathbb{Z} \times \mathbb{Z}/p^s\mathbb{Z})/\langle (1,1) \rangle \cong \mathbb{Z}/p^m\mathbb{Z}$, where the last isomorphism is $(a,b) \mapsto a - b \bmod p^m$.
- *Compositum*: $\langle g \rangle \cap \ker(G \to \mathrm{Gal}(K/\mathbb{Q})) = \{1\}$ because $g \mapsto \sigma$ has order $p^s = |g|$. So $\mathrm{Gal}(A/K)$ and $\mathrm{Gal}(A/F) = \langle g \rangle$ together generate G, giving $KF = A$. Since $F \cap \Lambda = \mathbb{Q}$ (q is unramified in F but totally ramified in Λ), we have $[F\Lambda : \mathbb{Q}] = p^{m+s} = [A:\mathbb{Q}]$, so $F\Lambda = A = K\Lambda$.
- *Unramified at q*: The inertia group of $A/\mathbb{Q}$ at q equals $\langle g \rangle$ (the ramification index of $A/\mathbb{Q}$ at q is p^s, achieved by g). Since g fixes $F = A^{\langle g \rangle}$, the inertia group maps trivially to $\mathrm{Gal}(F/\mathbb{Q})$, so q is unramified in $F/\mathbb{Q}$. $\square$

Lemma 52. *Every extension of $\mathbb{Q}$ of degree greater than 1 is ramified at some finite prime.*

Proof. $\square$

Proposition 53. *The only quadratic extensions of $\mathbb{Q}$ unramified outside 2 are $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-2})$, and $\mathbb{Q}(\sqrt{2})$. In particular, the maximal real abelian 2-extension of $\mathbb{Q}$ with exponent 2 and unramified outside 2 is $\mathbb{Q}(\sqrt{2})$, which is the subfield of $\mathbb{Q}(\zeta_8)$ of degree 2.*

Proof. A quadratic extension of $\mathbb{Q}$ unramified outside 2 has discriminant a power of 2, so it is of the form $\mathbb{Q}(\sqrt{d})$ with $d \in \{-1, \pm 2\}$. This gives exactly the three fields listed. Among these, only $\mathbb{Q}(\sqrt{2})$ is real. $\square$

Proposition 54. *Let $K/\mathbb{Q}$ be a cyclic extension of degree 2^m unramified outside 2. Then K is cyclotomic.*

Proof. We proceed by induction on m. The base case $m = 1$ is 53. Assume $m \geq 2$ and the result holds for degree 2^{m-1}.

If K is nonreal, complex conjugation $c \in \mathrm{Gal}(K/\mathbb{Q})$ is the unique element of order 2 in the cyclic group $\mathrm{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/2^m\mathbb{Z}$, and $M = K^c$ is totally real, cyclic of degree 2^{m-1} over $\mathbb{Q}$, and unramified outside 2. By induction, $M \subseteq \mathbb{Q}(\zeta_n)$ for some n. The extension K/M is quadratic and unramified outside 2, so $K = M(\sqrt{d})$ with $d \in \{-1, \pm 2\}$; in every case $K \subseteq M(\zeta_8) \subseteq \mathbb{Q}(\zeta_{8n})$, so K is cyclotomic.

Let K' be the maximal real subfield of $\mathbb{Q}(\zeta_{2^{m+2}})$; it is cyclic of degree 2^m over $\mathbb{Q}$ and unramified outside 2. Suppose $K'K/\mathbb{Q}$ is not cyclic. Then $\mathrm{Gal}(K'K/\mathbb{Q})$ is a non-cyclic 2-group, so it has $(\mathbb{Z}/2\mathbb{Z})^2$ as a quotient, giving three distinct quadratic subfields of $K'K$. All are unramified outside 2. Since $K'K$ is totally real (both K and K' are), all three quadratic subfields are real. But by 53 there is only one real quadratic field unramified outside 2, namely $\mathbb{Q}(\sqrt{2})$, a contradiction. Therefore $K'K/\mathbb{Q}$ is cyclic, and 50 gives $K \subseteq K' \subseteq \mathbb{Q}(\zeta_{2^{m+2}})$. □

Proposition 55. *Let $K/\mathbb{Q}$ be a cyclic extension of odd prime power degree p^m unramified outside p. Then K is cyclotomic.*

Proof. Let K' be the unique subfield of degree p^m in $\mathbb{Q}(\zeta_{p^{m+1}})$; it is cyclic of degree p^m and unramified outside p. Suppose $K'K/\mathbb{Q}$ is not cyclic. Then $\mathrm{Gal}(K'K/\mathbb{Q})$ is a non-cyclic p-group, so it has $(\mathbb{Z}/p\mathbb{Z})^2$ as a quotient, giving a subfield $E \subseteq K'K$ with $\mathrm{Gal}(E/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^2$. Since K and K' are unramified outside p, so is E. But by 56 the maximal abelian extension of exponent p unramified outside p is cyclic of degree p, so no extension with Galois group $(\mathbb{Z}/p\mathbb{Z})^2$ exists. This contradiction shows $K'K/\mathbb{Q}$ is cyclic, so 50 gives $K \subseteq K' \subseteq \mathbb{Q}(\zeta_{p^{m+1}})$. □

5.2 The prime degree case

Proposition 56. *Let p be an odd prime. The maximal abelian extension of $\mathbb{Q}$ of exponent p unramified outside p is cyclic of degree p: it is the subfield of degree p of $\mathbb{Q}(\zeta_{p^2})$.*

Proof. Let $K/\mathbb{Q}$ be a cyclic extension of prime degree p unramified outside p. Set $F = \mathbb{Q}(\zeta_p)$ and $L = KF$.

1. By 57, $L = F(\sqrt[p]{\mu})$ for some $\mu \in \mathcal{O}_F$, $\mu \neq 0$, with $X^p - \mu$ irreducible over F.
2. By 60, $p \mid v_{\mathfrak{q}}(\mu)$ for every prime $\mathfrak{q}$ of $\mathcal{O}_F$.
3. By 62, $(\mu) = \mathfrak{a}^p$ for some ideal $\mathfrak{a}$ of $\mathcal{O}_F$.
4. By 63, $\mathfrak{a}$ is principal.
5. By 64, $\mu = \alpha^p \eta$ for some $\alpha \in \mathcal{O}_F$ and unit $\eta \in \mathcal{O}_F^\times$.
6. By 65, $L = \mathbb{Q}(\zeta_{p^2})$.

Therefore $K \subseteq L = \mathbb{Q}(\zeta_{p^2})$. The Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_{p^2})/\mathbb{Q}) \cong (\mathbb{Z}/p^2\mathbb{Z})^\times \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z}$ has a unique subgroup of index p, giving a unique subfield of degree p over $\mathbb{Q}$ in $\mathbb{Q}(\zeta_{p^2})$. Since $K \cap F = \mathbb{Q}$ (as $\gcd([K : \mathbb{Q}], [F : \mathbb{Q}]) = \gcd(p, p-1) = 1$) and $[\mathbb{Q}(\zeta_{p^2}) : F] = p$, K is exactly this unique subfield. The same applies to any other cyclic extension of degree p unramified outside p, so the maximal such abelian extension of exponent p is cyclic of degree p. □

The proof of 56 proceeds via the following lemmas. Let $K/\mathbb{Q}$ be a cyclic extension of prime degree p unramified outside p. Set $F = \mathbb{Q}(\zeta_p)$ and $L = KF$.

Lemma 57. *With the above notation, $L = F(\sqrt[p]{\mu})$ for some nonzero $\mu \in \mathcal{O}_F$, and L/F is unramified outside p.*

Proof. Since $[K : \mathbb{Q}] = p$ and $[F : \mathbb{Q}] = p - 1$ are coprime, $K \cap F = \mathbb{Q}$, so $[L : F] = [KF : F] = [K : \mathbb{Q}] = p$. The extension L/F is Galois with cyclic Galois group of order p, and F contains the p-th roots of unity μ_p. By Kummer theory (`IsCyclic` + `Kummer`), $L = F(\sqrt[p]{\mu})$ for some $\mu \in F^\times$. Multiplying by a p-th power if necessary, we may assume $\mu \in \mathcal{O}_F$.

For the ramification: let $q \neq p$ be a prime. Since $K/\mathbb{Q}$ is unramified at q and $F/\mathbb{Q}$ is unramified at q (as $F = \mathbb{Q}(\zeta_p)$ is only ramified at p), the compositum $L = KF$ is also unramified at q. □

Lemma 58. *Since $L/\mathbb{Q}$ is Galois, for every $\sigma_a \in \mathrm{Gal}(F/\mathbb{Q})$ there exists $\xi \in F^\times$ and $a \in \mathbb{Z}$ such that $\sigma_a(\mu) = \xi^p \mu^a$.*

Proof. This is the backward direction of `isGalois_iff_forall_apply_eq_pow_mul_zpow`: since $L/\mathbb{Q}$ is Galois (which holds because $L = KF$ is the compositum of the abelian extensions $K/\mathbb{Q}$ and $F/\mathbb{Q}$), the criterion gives the required ξ and a for every $\sigma_a \in \mathrm{Gal}(F/\mathbb{Q})$. □

Lemma 59. *Let $\mathfrak{q}$ be a prime ideal of F with $(\mu) = \mathfrak{q}^r\mathfrak{a}$ with $\mathfrak{q} \nmid \mathfrak{a}$. If $p \nmid r$ and $L/\mathbb{Q}$ is abelian, then $\mathfrak{q}$ splits completely in $F/\mathbb{Q}$.*

Proof. Let σ_a be an element of the decomposition group of $\mathfrak{q}$ in $\mathrm{Gal}(F/\mathbb{Q})$, so $\sigma_a(\mathfrak{q}) = \mathfrak{q}$. Since $L/\mathbb{Q}$ is abelian, by 58 there exists $\xi \in F^\times$ such that $\sigma_a(\mu) = \xi^p \mu^a$. Since $\sigma_a(\mathfrak{q}) = \mathfrak{q}$, applying σ_a to the factorization $(\mu) = \mathfrak{q}^r\mathfrak{a}$ gives $(\xi^p\mu^a) = \mathfrak{q}^r\sigma_a(\mathfrak{a})$, hence $\mathfrak{q}^r \| \xi^p\mu^a$. But also $(\mu^a) = \mathfrak{q}^{ar}\mathfrak{a}^a$, so $\mathfrak{q}^r \| \xi^p\mu^a$ implies $r \equiv ar \pmod{p}$. Since $p \nmid r$, this forces $a \equiv 1 \pmod{p}$, so $\sigma_a = 1$ and $\mathfrak{q}$ splits completely in $F/\mathbb{Q}$. □

Lemma 60. *For any prime $\mathfrak{q}$ of $\mathcal{O}_F$, $p \mid v_\mathfrak{q}(\mu)$.*

Proof. Write $r = v_\mathfrak{q}(\mu)$.

Case 1: $\mathfrak{q} \nmid p$. By 57, L/F is unramified at $\mathfrak{q}$, so $e(\mathfrak{Q}|\mathfrak{q}) = 1$ for any prime $\mathfrak{Q}$ of L above $\mathfrak{q}$. Let $\alpha \in L$ satisfy $\alpha^p = \mu$. Then

$$p \cdot v_\mathfrak{Q}(\alpha) = v_\mathfrak{Q}(\alpha^p) = v_\mathfrak{Q}(\mu) = e(\mathfrak{Q}|\mathfrak{q}) \cdot v_\mathfrak{q}(\mu) = v_\mathfrak{q}(\mu) = r,$$

so $p \mid r$.

Case 2: $\mathfrak{q} = \mathfrak{p} = (1-\zeta_p)$. Since $F/\mathbb{Q}$ is totally ramified at p, $\mathfrak{p}$ is the unique prime of F above p, so its decomposition group in $\mathrm{Gal}(F/\mathbb{Q})$ is all of $\mathrm{Gal}(F/\mathbb{Q})$. Since $|\mathrm{Gal}(F/\mathbb{Q})| = p - 1 > 1$, $\mathfrak{p}$ does not split completely in $F/\mathbb{Q}$. By the contrapositive of 59, $p \mid r$. □

Lemma 61. *There exists $\nu \in \mathcal{O}_F$, $\nu \neq 0$, such that $L = F(\sqrt[p]{\nu})$, $X^p - \nu$ is irreducible over F, and $(1 - \zeta_p) \nmid \nu$.*

Proof. By 60, $p \mid v_\mathfrak{p}(\mu)$; write $v_\mathfrak{p}(\mu) = pk$. Set $\nu = \mu/(1 - \zeta_p)^{pk} \in \mathcal{O}_F$. Since $(1 - \zeta_p)^k \in F^\times$, we have $F(\sqrt[p]{\nu}) = F(\sqrt[p]{\mu}/(1 - \zeta_p)^k) = F(\sqrt[p]{\mu}) = L$ (by 57). By construction $v_\mathfrak{p}(\nu) = 0$, i.e. $(1 - \zeta_p) \nmid \nu$. □

Lemma 62. $(\mu) = \mathfrak{a}^p$ *for some ideal $\mathfrak{a}$ of $\mathcal{O}_F$.*

Proof. By 60, $p \mid v_\mathfrak{q}(\mu)$ for every prime $\mathfrak{q}$. Therefore $(\mu) = \mathfrak{a}^p$ where $\mathfrak{a} = \prod_\mathfrak{q} \mathfrak{q}^{v_\mathfrak{q}(\mu)/p}$. □

Lemma 63. *The ideal class* $[\mathfrak{a}]$ *is trivial, i.e.,* $\mathfrak{a}$ *is principal.*

Proof. From $(\mu) = \mathfrak{a}^p$ and $\sigma_a(\mu) = \xi^p \mu^a$ (by 58), we get $\sigma_a(\mathfrak{a})^p = \sigma_a(\mathfrak{a}^p) = \sigma_a((\mu)) = (\xi^p \mu^a) = (\xi)^p \cdot \mathfrak{a}^{pa}$. Since the ideal monoid is torsion-free, $\sigma_a(\mathfrak{a}) = (\xi) \cdot \mathfrak{a}^a$, hence $\sigma_a(c) = c^a$ for the ideal class $c = [\mathfrak{a}]$ and every $1 \le a < p$.

Note that we work throughout with the *class* c: unlike in the per-ideal form of Stickelberger, we never need $\mathfrak{a}$ itself to be coprime to p or to have odd norm. By 45 applied with $m = p$, the class c is annihilated by the Stickelberger element $\Theta = \sum_{a=1}^{p-1} a\sigma_a^{-1} \in \mathbb{Z}[G]$. We compute c^Θ:

$$1 = c^\Theta = \prod_{a=1}^{p-1} \sigma_a^{-1}(c)^a = \prod_{a=1}^{p-1} c^{a^{-1} \cdot a} = c^{p-1},$$

where we used $\sigma_a^{-1}(c) = \sigma_{a^{-1}}(c) = c^{a^{-1}}$, with a^{-1} the inverse of a modulo p. Since $(\mu) = \mathfrak{a}^p$, the ideal $\mathfrak{a}^p$ is principal, so $c^p = 1$ and hence $c^{p-1} = c^{-1}$. Combining with $c^\Theta = 1$ gives $c^{-1} = 1$, i.e. $c = [\mathfrak{a}] = 1$ and $\mathfrak{a}$ is principal. $\square$

Lemma 64. *There exists a unit* $\eta \in \mathcal{O}_F^\times$ *such that* $\mu = \alpha^p \eta$ *for some* $\alpha \in \mathcal{O}_F$.

Proof. By 63, $\mathfrak{a} = (\alpha)$ for some $\alpha \in \mathcal{O}_F$. Then $(\mu) = \mathfrak{a}^p = (\alpha^p)$, so $\mu/\alpha^p \in \mathcal{O}_F^\times$. Setting $\eta = \mu/\alpha^p$ gives $\mu = \alpha^p \eta$. $\square$

Lemma 65. *The unit* η *is a* p*-th power times a root of unity, so* $\mu = \zeta_p^t$ *for some* t*. Therefore* $L = \mathbb{Q}(\zeta_{p^2})$.

Proof. Let σ_{-1} denote complex conjugation on F, so $\sigma_{-1}(\zeta_p) = \zeta_p^{-1}$. Every unit of the CM field F is a root of unity times a unit of its maximal real subfield, so write $\eta = \zeta_p^{t_0} \varepsilon$ with $\sigma_{-1}(\varepsilon) = \varepsilon$. With $\mu = \alpha^p \eta$, squaring gives

$$\mu^2 = \alpha^{2p} \zeta_p^t \varepsilon^2, \qquad t := 2t_0.$$

Since $L/\mathbb{Q}$ is abelian, 58 applied to σ_{-1} gives $\xi \in F^\times$ with $\sigma_{-1}(\mu) = \xi^p \mu^{-1}$, hence $\sigma_{-1}(\mu^2) = \xi^{2p} \mu^{-2}$. Applying σ_{-1} to the displayed identity,

$$\sigma_{-1}(\alpha)^{2p} \zeta_p^{-t} \varepsilon^2 = \xi^{2p} \alpha^{-2p} \zeta_p^{-t} \varepsilon^{-2}.$$

Cancelling ζ_p^{-t} gives

$$\varepsilon^4 = \left(\frac{\xi}{\alpha \, \sigma_{-1}(\alpha)} \right)^{2p},$$

so ε^4 is a p-th power in $F^\times$. As p is odd, $\gcd(4, p) = 1$, so ε is a p-th power, say $\varepsilon = \delta^p$ with $\delta \in \mathcal{O}_F^\times$. Hence $\mu = \alpha^p \zeta_p^{t_0} \varepsilon = (\alpha\delta)^p \zeta_p^{t_0}$ with $\alpha\delta \in \mathcal{O}_F$, and

$$L = F(\sqrt[p]{\mu}) = F(\sqrt[p]{\zeta_p^{t_0}}) \subseteq \mathbb{Q}(\zeta_{p^2}).$$

Since $[L : F] = p = [\mathbb{Q}(\zeta_{p^2}) : F]$, we conclude $L = \mathbb{Q}(\zeta_{p^2})$. $\square$

Chapter 6

Auxiliary results

Lemma 66. *The ramification index of* $\mathfrak{P}$ *above* p *is* $p-1$*, i.e.,*

$$e(\mathfrak{P} \mid p) = p - 1.$$

Proof. This follows from the general formula for the ramification index in cyclotomic extensions. □

Lemma 67. *We have* $\zeta_p - 1 \notin \mathfrak{P}^2$*.*

Proof. By 66, $e(\mathfrak{P} \mid p) = p - 1$. Since ζ_p is a primitive p-th root of unity, $(\zeta_p - 1)$ generates the unique prime above p in $\mathbb{Q}(\zeta_p)$, which has ramification index $p - 1$. Hence $v_{\mathfrak{P}}(\zeta_p - 1) = 1$, so $\zeta_p - 1 \notin \mathfrak{P}^2$. □

Lemma 68. *For any additive character* ψ *of* $\mathcal{O}_K/P$*,*

$$g(1, \psi) = \begin{cases} q - 1 & \textit{if } \psi = 1 \\ -1 & \textit{otherwise,} \end{cases}$$

where $\psi = 1$ *denotes the trivial character (identically equal to* 1*).*

Proof. □